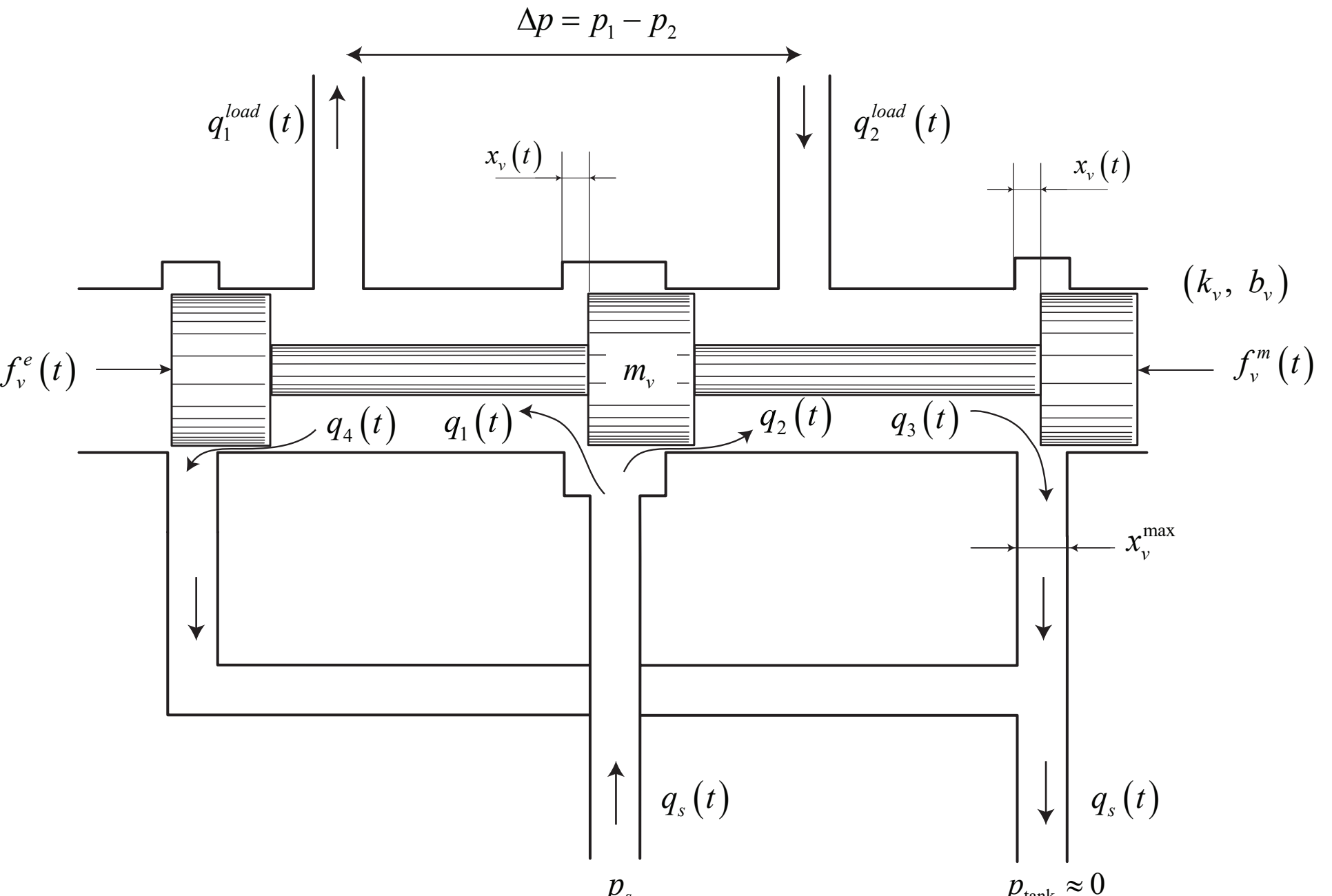

$\Delta p = p_1 - p_2$
$q_1^{load}(t)$
$q_2^{load}(t)$
$x_v(t)$
$x_v(t)$
$(k_v,\ b_v)$
$f_v^e(t)$
m_v
$f_v^m(t)$
$q_4(t)$
$q_1(t)$
$q_2(t)$
$q_3(t)$
$x_v^{\max}$
$q_s(t)$
$q_s(t)$
p_s
$p_{\text{tank}} \approx 0$